

- 4 (a) A 50 W transmitter emits microwaves of frequency 10 GHz in all directions, as shown in Fig. 4.1. A receiver is initially at point O, which is 500 m due east of the transmitter, and receives microwaves at an amplitude of 15 cm. The receiver is then moved 300 m southwards to point P.

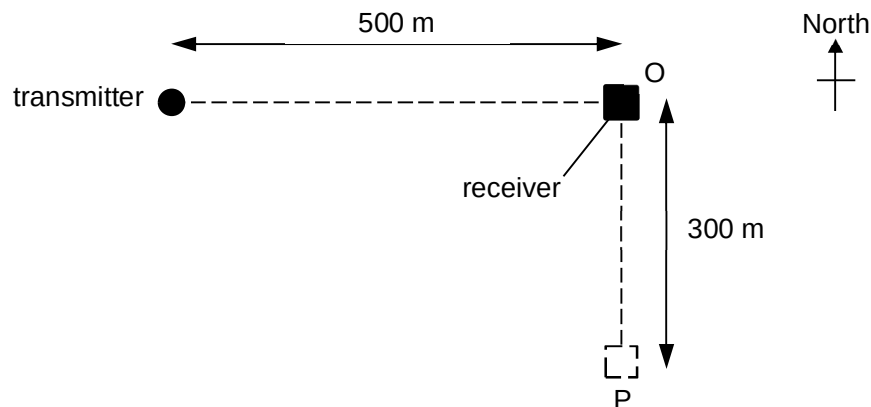


Fig. 4.1

- (i) Determine the wavelength of the microwaves.

wavelength = m [1]

- (ii) Determine the intensity of the microwaves at P.

intensity = W m^{-2} [2]

- (iii) Determine the amplitude of the microwaves detected by the receiver at P.

amplitude = cm [2]

- (b) Fig. 4.2 shows microwaves passing through a series of metal plates with narrow slits. The microwaves are polarised in the plane parallel to the length of the slit after passing through each metal plate. The metal plates are arranged such that the slits in plates A and C are parallel and perpendicular to the axis as shown in Fig. 4.2. The slit in plate B is placed at an angle θ from the axis. The intensities of the microwaves after passing through plates A and C are I_0 and I respectively.

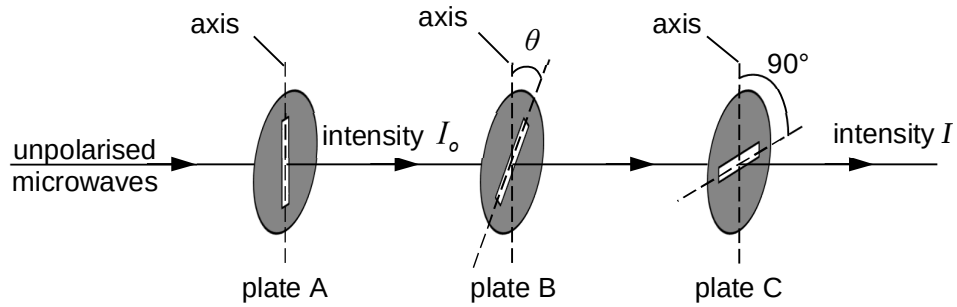


Fig. 4.2

Show that $I = I_0 \cos^2 \theta \sin^2 \theta$.

[3]

5 Two charged metal spheres A and B are situated in a vacuum. The distance between the